

Advanced Data Structures

Summer semester 2026

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2026-04-20 Mon 14:00

Organizational matters

Lectures:

- In English
 - Monday 14:00 - 15:30
 - Building 50.34, room 236
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Grade:

- 80% oral exam, 20 minutes
- 20% programming project
 - Details coming a few weeks
 - **Requires registration**

Materials

This year's materials:

- The course page: ae.itk.kit.edu/english/5028.php
- Ragnar's blog: curiouscoding.nl/teaching
 - Lecture notes and/or slides available shortly before the lecture

Previously taught by [Florian Kurpicz](#):

- [SS 2024 course materials & slides](#) (in English)
- Recorded lectures on YouTube ([@kurpicz](#), in German):
 - [SS 2022](#)
 - [SS 2023](#)

Related MIT Advanced Data Structures course by Erik Demaine:

- [Course overview](#)
- [2012 recordings](#)

Tentative course overview

Ragnar (5 weeks):

- Today: Models of computation
- Rank & Select
- Applications of Rank & Select:
 - Elias Fano
 - Succinct trees
 - Succinct planar graphs
- Range minimum queries
 - Topic of **practicum**
- Segment Trees & Fenwick Trees

Tentative course overview

Stefan Walzer & Stefan Hermann (9 weeks):

- Sorted sequences / predecessor structures
 - Van Emde Boas tree
 - x-Fast trie
 - y-Fast trie
- External memory model
 - Cache-oblivious B-trees
- Learned indexes
- Streaming and Sketching algorithms
- Splay trees
- Modern hash tables

Social time!

First slides: **Models of computation**
